

Proxy-Informed Gaussian Process Priors for Bayesian Optimisation of Very Expensive Objectives

James Avery

Department of Computer Science, University of Copenhagen
avery@di.ku.dk

June 2026 — working draft

Abstract

Bayesian optimisation (BO) of very expensive objective functions — e.g. coupled-cluster electronic structure energies that take months per evaluation — typically starts from an uninformative Gaussian process (GP) prior and must spend a large fraction of its small evaluation budget rediscovering the gross structure of the objective. In many such applications a fast surrogate of the objective exists (harmonic approximations in microseconds, semi-empirical methods in seconds) together with a *known pointwise bound* on its error: the true objective deviates from the proxy by at most a known percentage, even though the locations of their optima may differ. We show how to encode this knowledge exactly where it belongs in the BO machinery: the GP prior mean is set to the proxy, and the prior covariance is scaled heteroscedastically so that the prior standard deviation at every point is the promised deviation band. The bound is treated as a hard upper limit on the prior amplitude — a single trainable scale factor may tighten the band as data accumulates but is constrained never to exceed it. The construction composes with arbitrary stationary correlation kernels, survives the data renormalisation used in practical BO implementations, and is available in the open-source package `veropt`. On the six-dimensional Hartmann benchmark with a proxy distorted by at most 1%, the proxy-informed prior comes within 0.2% of the global optimum after a single Bayesian batch (12 evaluations) in every repetition, whereas a flat prior is still 18% short of the optimum when the budget of 24 evaluations is exhausted.

1 Introduction

Bayesian optimisation is the method of choice for optimising black-box functions whose evaluations are so expensive that only a small number —

often around one hundred — can be afforded (Shahriari et al., 2016; Frazier, 2018). A Gaussian process surrogate is fitted to the evaluations made so far, and an acquisition function balances exploration against exploitation to select the next batch of points.

The standard GP prior is maximally uninformative: a constant mean and a stationary kernel with a single global amplitude. Every evaluation must then contribute to learning both the gross structure of the objective *and* the fine structure near the optimum. For the most expensive objectives this is wasteful, because the gross structure is often already known. Our motivating example is molecular electronic structure: a CCSD(T)-level energy surface may cost months of computation per point, while a harmonic approximation to the same surface is available in microseconds and semi-empirical electronic structure methods in seconds. Crucially, these cheap surrogates frequently come with *quantitative* error guarantees of the form “the true energy is within 1% of the approximation, everywhere”. The minima of the cheap and the expensive surface need not coincide — otherwise the problem would be solved — but pointwise, the expensive function is trapped in a known band around the cheap one.

This paper describes a simple, well-founded way to inject exactly this kind of knowledge into BO: build the GP prior from the proxy. The prior mean is the proxy itself, and the prior covariance is rescaled so that the prior standard deviation at each point equals the promised band. The result is that the surrogate starts out already knowing everything the proxy knows, and the expensive evaluations are spent exclusively on learning the *residual* between truth and proxy — a function whose amplitude is bounded by the guarantee. Related ideas appear in multi-fidelity BO (Kennedy and O’Hagan, 2000; Forrester et al., 2007), where cheap and expensive evaluations are modelled jointly, and more broadly in grey-box BO (Astudillo and Frazier, 2021), which exploits known structure of the objective. The variant presented here is deliberately minimal: it requires no evaluations of the expensive function to fit the relationship between fidelities — the auto-regressive coefficient and discrepancy process of Kennedy and O’Hagan (2000) are replaced by an identity coupling and an amplitude bound, both supplied a priori by the user’s guarantee. This matters precisely in the regime we target: with months per evaluation, there is no budget to learn the cross-fidelity relationship empirically.

2 Method

2.1 Setting

Let $f: X \subset \mathbb{R}^d \rightarrow \mathbb{R}$ be the expensive objective, to be maximised under a budget of $N \sim 10^2$ evaluations. Let $g: X \rightarrow \mathbb{R}$ be a fast proxy (μ s per

evaluation), and assume the pointwise guarantee

$$|f(x) - g(x)| \leq \varepsilon(x) \quad \text{for all } x \in X, \quad (1)$$

with ε known. The two cases of practical interest are the *relative* bound $\varepsilon(x) = r |g(x)|$ (“ f is within 1% of g ”, $r = 0.01$) and the *absolute* bound $\varepsilon(x) = \varepsilon_0$. Nothing is assumed about the locations of the optima of f and g .

2.2 The proxy-informed prior

We place on f the Gaussian process prior

$$f \sim \mathcal{GP}(m, k), \quad m(x) = g(x), \quad k(x, x') = c^2 \sigma(x) \sigma(x') \rho(x, x'), \quad (2)$$

where ρ is a correlation kernel ($\rho(x, x) = 1$) and

$$\sigma(x) = \frac{\varepsilon(x)}{\kappa}, \quad (3)$$

with $\kappa > 0$ a confidence parameter discussed below. The kernel in (2) is the standard non-stationary scaling of a correlation kernel by an outer product of positive functions, and is therefore a valid (positive semi-definite) covariance. In practice ρ is obtained from an arbitrary base kernel k_b by normalising with its diagonal,

$$\rho(x, x') = \frac{k_b(x, x')}{\sqrt{k_b(x, x) k_b(x', x')}}, \quad (4)$$

so that the construction stays calibrated even for base kernels with $k_b(x, x) \neq 1$ — kernel sums and spectral mixtures being the common offenders: a sum of two Matérn kernels has $k_b(x, x) = 2$ and would otherwise silently inflate the band by $\sqrt{2}$ past the promised bound.

Equation (2) is most easily read through the residual $h = f - g$: the prior states $h \sim \mathcal{GP}(0, k)$ with pointwise standard deviation $c \sigma(x)$, i.e. the model believes the truth deviates from the proxy by a smooth function whose typical size at x is a known fraction of the band $\varepsilon(x)$. With $c = 1$ and $\kappa = 2$ the prior places $\sim 95\%$ of its mass inside the band (1) at every point; $\kappa = 3$ gives $\sim 99.7\%$. The bound is thus encoded *softly* — a GP cannot represent a hard truncation — and κ expresses how literally the guarantee is taken. We default to $\kappa = 2$.

2.3 The bound as a hard cap on a trained amplitude

The guarantee (1) is an *upper* bound: the proxy may well be better than its worst case. We therefore make the global amplitude factor c in (2) a

hyperparameter, trained alongside the kernel lengthscales by type-II maximum likelihood, but constrained to

$$c \in (c_{\min}, 1], \quad (5)$$

implemented by a sigmoidal reparametrisation (a `gpytorch` (Gardner et al., 2018) interval constraint). The cap at 1 guarantees that no amount of hyperparameter optimisation can inflate the prior beyond the promised band, while the marginal likelihood is free to *tighten* the band as the data reveal that the proxy is more accurate than worst-case — which immediately sharpens the posterior everywhere, not only near evaluations. The floor $c_{\min}$ (default 0.1) prevents overconfident collapse from the handful of points available early on. Setting c constant at 1 recovers the literal worst-case prior.

Two practical guards are worth stating. First, under a relative bound $\sigma(x) \rightarrow 0$ wherever $g(x) \rightarrow 0$, which pins the GP to the proxy and can destabilise Cholesky factorisations; a small floor $\varepsilon(x) \leftarrow \max(\varepsilon(x), \varepsilon_{\text{floor}})$ restores numerical safety. Second, since $\rho(x, x) = 1$ exactly by (4), the prior variance at any point is $c^2 \sigma(x)^2$ by construction — there is no additional global outpuscale to interfere with the calibration.

2.4 Interaction with data normalisation

Practical BO implementations whiten their data: inputs and observed objective values are affinely normalised (zero mean, unit variance per dimension), and the normalisers are refitted as new evaluations arrive. The proxy, by contrast, is defined in physical units. The prior mean must therefore be evaluated as

$$\tilde{m}(\tilde{x}) = T_y(g(T_x^{-1}(\tilde{x}))), \quad (6)$$

where T_x, T_y are the *current* input and output normalisations and tildes denote whitened quantities. The band, being the standard deviation of a *deviation* rather than a value, transforms through the scale of T_y only:

$$\tilde{\sigma}(\tilde{x}) = T_y(\sigma(x)) - T_y(0) = \sigma(x)/s_y, \quad (7)$$

with s_y the output scale — applying the full affine map here would be incorrect, silently shifting the band by the normalisation offset. Because the normalisers change between refits, the implementation binds them to the mean and kernel by late-bound closures that read the current normaliser at call time.

2.5 Cost

The proxy is evaluated at every posterior query, which during acquisition optimisation (here: dual annealing with sequential proximity-punished batch selection) means thousands of calls per suggested batch. Microsecond- to

millisecond-scale proxies are negligible; second-scale proxies (semi-empirical methods) dominate the suggestion step and benefit from memoisation, which we leave as future work. Against an objective costing days to months per evaluation, even the un-memoised cost is irrelevant.

3 Implementation

The method is implemented in the open-source Bayesian optimisation package `veropt` (Stoustrup et al., 2026), which was built for tuning the VEROS ocean simulator (Häfner et al., 2018) and targets exactly the expensive-objective regime considered here. The user subclasses a savable `ProxyMeanFunction` (physical units in, physical units out) and passes it through the model configuration:

```

optimiser = bayesian_optimiser(
    n_initial_points=8,
    n_bayesian_points=32,
    n_evaluations_per_step=4,
    objective=expensive_objective,
    model={
        'kernels': 'matern',
        'proxy_prior': my_proxy,
        'proxy_prior_settings': {'bound_value': 0.01},
    },
    n_points_before_fitting=4,
)

```

Internally, a `ProxyMean` module supplies the mean of (2), and a `ProxyScaledKernel` wraps the chosen correlation kernel with the $c^2\sigma(x)\sigma(x')$ factor; the construction composes with every kernel offered by the package and with its per-step renormalisation, batch acquisition, and state serialisation. In the multi-objective case a proxy may be supplied per objective, with uninformed objectives left at the default prior.

4 Experiments

We use the six-dimensional Hartmann function (maximised, optimum $f^* = 3.3224$) as the “expensive” objective and construct a proxy by multiplicative harmonic distortion,

$$g(x) = f(x) (1 + 0.01 \sin(6\pi x_1)), \quad (8)$$

so that the relative bound (1) with $r = 0.01$ holds exactly, while the distortion shifts the local geometry of the optimum. Both methods use a Matérn-5/2 kernel with ARD, batches of 4, 8 initial random points followed by 16 Bayesian-phase points (a total budget of 24 evaluations), and `veropt`’s defaults otherwise.

Table 1: Best objective value found (mean $\pm$ standard deviation over five seeds) after n evaluations of the Hartmann-6 objective ($f^* = 3.3224$). Both methods share seeds, budget, kernel, and acquisition machinery; they differ only in the prior.

n	Flat prior	Proxy-informed prior
8 (random)	1.117 ± 0.644	1.117 ± 0.644
12	1.719 ± 0.451	3.3152 ± 0.0010
16	2.108 ± 0.387	3.3177 ± 0.0041
20	2.562 ± 0.247	3.3177 ± 0.0041
24	2.731 ± 0.276	3.3193 ± 0.0041

The proxy-informed runs use $\kappa = 2$, a trainable amplitude $c \in (0.1, 1]$, and an absolute band floor of $\varepsilon_{\text{floor}} = 0.01$. Each configuration is repeated five times with paired initial points (the torch seed is shared between arms; the acquisition optimiser retains its own stochasticity).

Figure 1 and Table 1 report the best objective value found as a function of the number of evaluations n , over five paired repetitions (the 8 initial random points are identical between the two arms). The effect of the prior is immediate: the *first* model-suggested point already lands at the global optimum, and after one Bayesian batch ($n = 12$) every proxy-informed run sits within 0.008 of the optimum — a relative optimality gap of 0.2%, i.e. at the scale of the proxy’s own error band — while the across-repetition spread collapses from ± 0.45 to ± 0.001 . The flat prior, by contrast, is still climbing at the end of the budget and terminates 0.59 below the optimum on average (an 18% gap) with nearly seventy times the spread. Reading the comparison in terms of evaluations rather than attained value: the proxy-informed optimiser needs 12 evaluations to surpass the level the flat prior has not reached after 24. For an objective costing months per evaluation, this is the difference between one and several years of computation. The trained amplitude factor is doing its intended work: across repetitions it settled at $c \in [0.69, 0.89]$, i.e. the marginal likelihood tightened the band 10–30% below the worst-case bound without ever being allowed to exceed it. We note the benchmark is deliberately favourable — the proxy is everywhere within 1% of the truth and the distortion is smooth — but this is precisely the regime the method is designed for, and the guarantee that makes it favourable is one the user must certify, not one the method assumes silently.

5 Limitations and future work

Soft bound. The hard guarantee (1) is encoded as a κ -sigma Gaussian band; at $\kappa = 2$, roughly 5% of prior mass violates the bound pointwise. Truncated

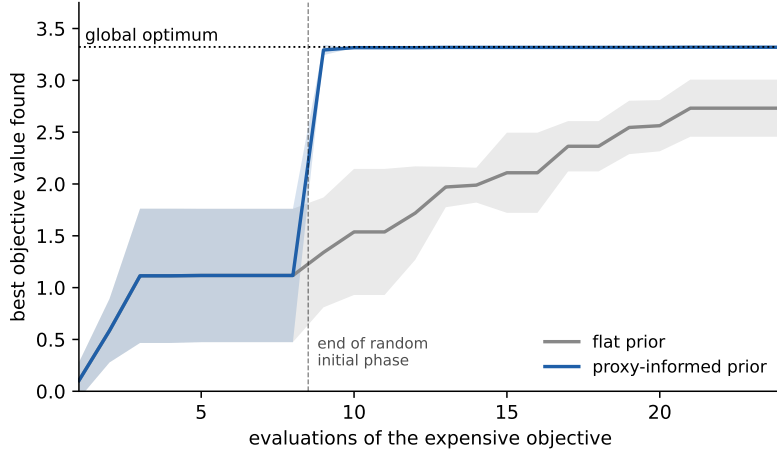


Figure 1: Best objective value found versus number of expensive evaluations on Hartmann-6 (mean over five paired repetitions; shaded bands are one standard deviation). The two arms share their 8 random initial points, so the curves coincide up to the dashed line. The first point suggested by the proxy-informed model already lands at the global optimum.

or constrained GPs would encode the bound exactly at considerably greater inferential cost.

Calibration of the residual smoothness. The bound fixes the *amplitude* of the residual but not its *lengthscale*; the latter is learned from data as usual. A proxy whose error oscillates faster than the learned lengthscale will be modelled overconfidently between evaluations.

Zero-data suggestion. With an informative prior the surrogate is meaningful before any expensive evaluation, so in principle the initial random phase could be replaced by acquisition over the prior alone. The current implementation still requires one batch of evaluations before the model is activated (its normalisers need at least two points); removing this restriction is mechanical but not yet done.

Gradients and memoisation. The proxy is currently evaluated without gradients — harmless under derivative-free acquisition optimisation, but a gradient-based acquisition optimiser would require differentiating through the prior mean. Memoisation of proxy calls would make second-scale proxies practical.

6 Conclusion

When an expensive objective comes with a cheap surrogate and a quantitative pointwise error guarantee — the everyday situation in computational chemistry and many other simulation sciences — that guarantee can be installed directly into the Gaussian process prior of a Bayesian optimiser: proxy as prior mean, error band as (heteroscedastic) prior standard deviation, bound as a hard cap on a trained amplitude. The construction is exact in its bookkeeping (valid PSD covariance, correct behaviour under data whitening), composes with standard kernels and acquisition machinery, and in benchmark experiments converts a months-per-point optimisation that would otherwise spend its budget rediscovering known structure into one that starts at the proxy and spends every evaluation on the residual.

Acknowledgements

The proxy-informed prior is implemented in `veropt`, developed by Aster Stoustrup and collaborators; this work builds directly on that foundation.

References

- R. Astudillo and P. I. Frazier. Thinking inside the box: A tutorial on grey-box Bayesian optimization. In *Proceedings of the 2021 Winter Simulation Conference*, 2021.
- A. I. J. Forrester, A. Söbester, and A. J. Keane. Multi-fidelity optimization via surrogate modelling. *Proceedings of the Royal Society A*, 463:3251–3269, 2007.
- P. I. Frazier. A tutorial on Bayesian optimization. *arXiv:1807.02811*, 2018.
- J. R. Gardner, G. Pleiss, D. Bindel, K. Q. Weinberger, and A. G. Wilson. GPyTorch: Blackbox matrix-matrix Gaussian process inference with GPU acceleration. In *Advances in Neural Information Processing Systems*, 2018.
- D. Häfner, R. L. Jacobsen, C. Eden, M. R. B. Kristensen, M. Jochum, R. Nuterman, and B. Vinter. Veros v0.1 – a fast and versatile ocean simulator in pure Python. *Geoscientific Model Development*, 11:3299–3312, 2018.
- M. C. Kennedy and A. O’Hagan. Predicting the output from a complex computer code when fast approximations are available. *Biometrika*, 87(1):1–13, 2000.

- B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016.
- A. Stoustrup et al. veropt: the versatile optimiser. <https://github.com/aster-stoustrup/veropt>, 2026.